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A Project Report

IMPLEMENTATION OF A SIEM SECURITY MONITORING SYSTEM USING WAZUH

Submitted by:

Mallak Al hakamani

- *Introduction:*

Security Information and Event Management (SIEM) plays a vital role in cybersecurity systems, where it is used to monitor and analyse events. It helps detect anomalies and potential security attacks in real time.

For this project, I developed a SIEM in a single virtual machine. It shows how security monitoring can be done in the system by gathering system logs and using them to identify interesting events. It also demonstrates the need to ensure the security of the system by monitoring it.

- *Objectives:*

1. To implement a SIEM system to monitor system events.
2. To collect and analyze system logs.
3. To detect anomalies using SIEM System.
4. To understand the importance of real-time monitoring in cybersecurity.

- *Tools and Technologies:*

I built the system using the following tools:

1. **VirtualBox**: to create and run the virtual machine.
2. **Ubuntu**: used as the operating system.
3. **Wazuh**: used as the security information and event management (SIEM) system.

- *About Wazuh:*

Wazuh is an open-source SIEM platform used for threat detection, log monitoring, and security event analysis. It helps security administrators monitor systems and respond to suspicious activities in real-time.

- *System Design:*

The SIEM system is based on a process:

1. Setup Stage: I installed a virtual machine using VirtualBox and installed Ubuntu on it.

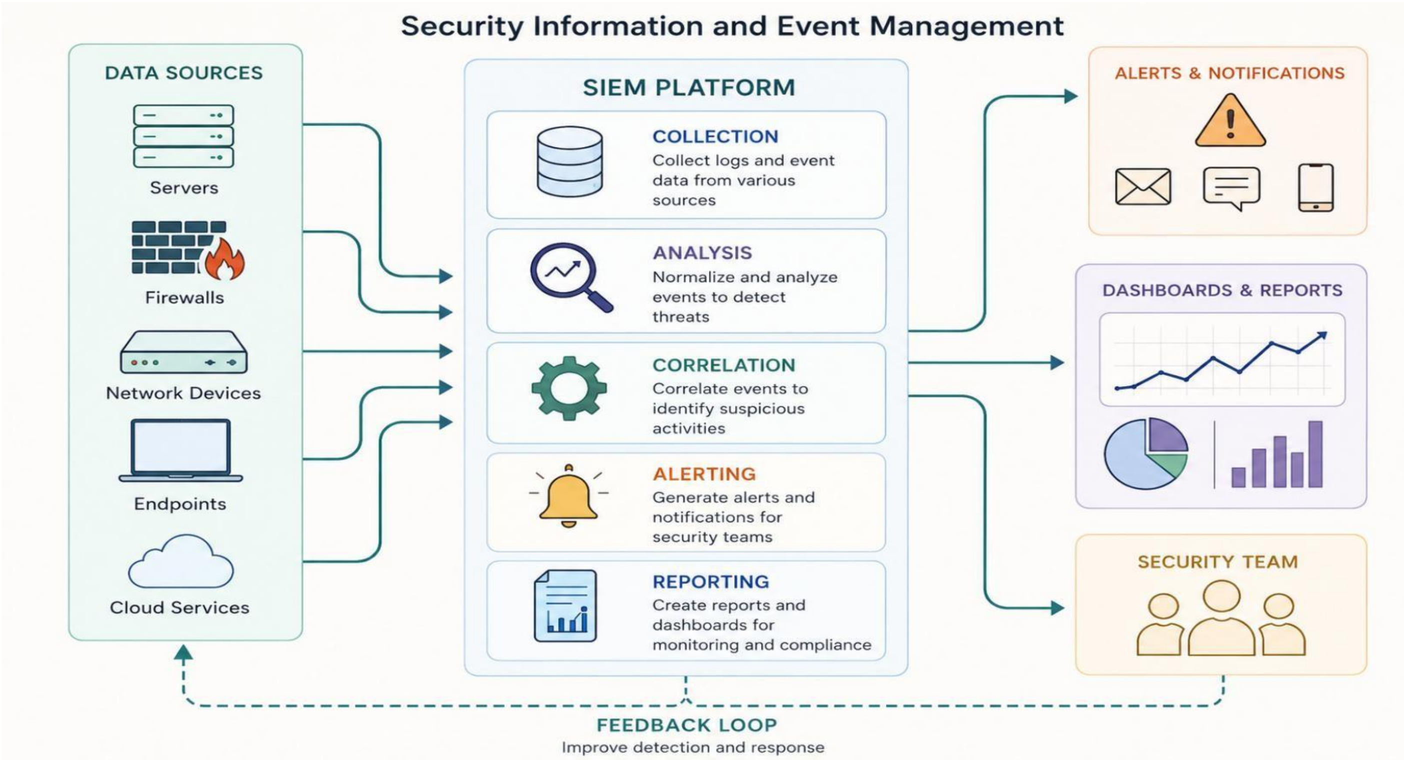
Then I installed and set up Wazuh as the SIEM solution.

2. **Log Collection Stage:** I set up the system to gather logs from the operating system, including logins and System activities.

3. Monitoring Stage: I used Wazuh to monitor and analyse these logs in real time to detect any anomalous activities.

4. Detection Stage (Output): The system then generated alerts for security concerns.

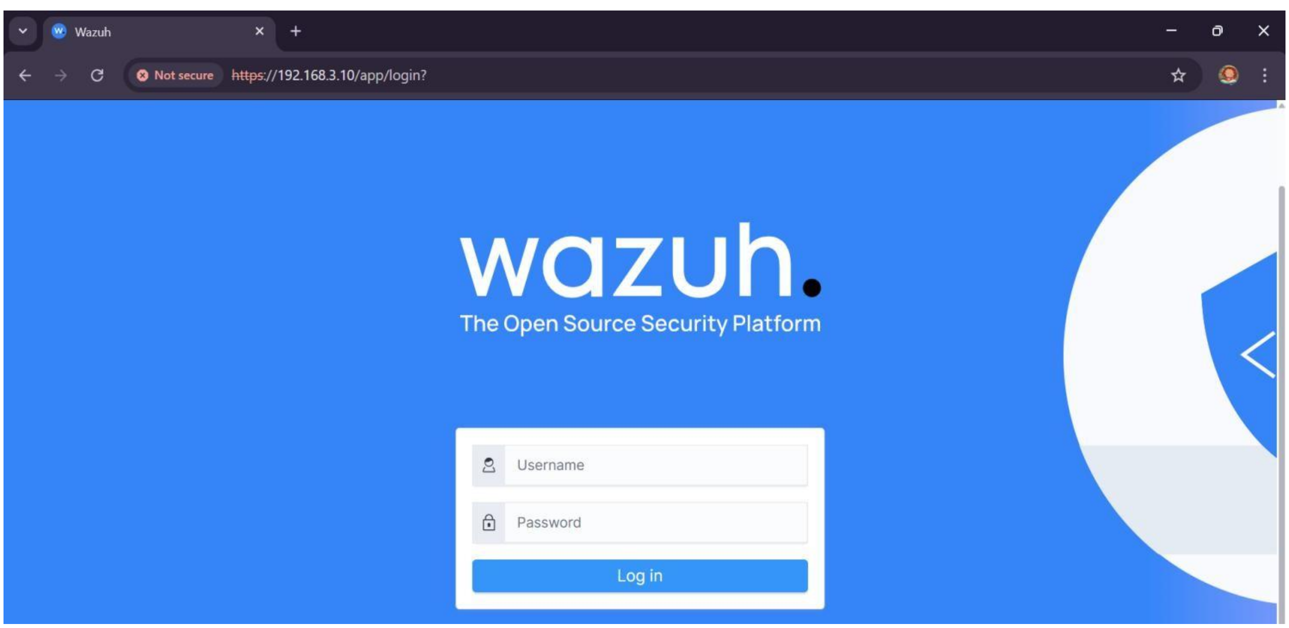
- ***Figure 1 :*** *SIEM System Architecture.*



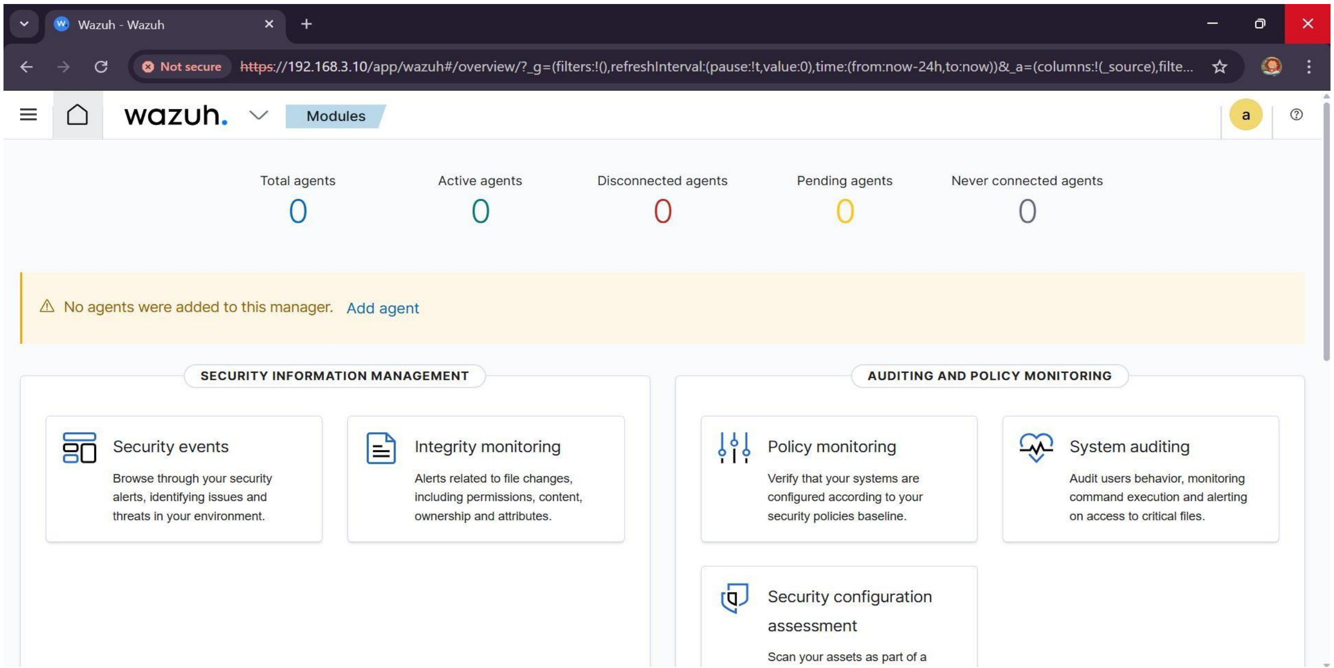
- ***Implementation:***

1. I have set up this system on a single virtual machine.
2. I installed Ubuntu on VirtualBox.
3. I set up Wazuh as the SIEM system.

- ***Figure 2*** : *Wazuh Login (user & password).*



- ***Figure 3** : Wazuh Dashboard Interface.*

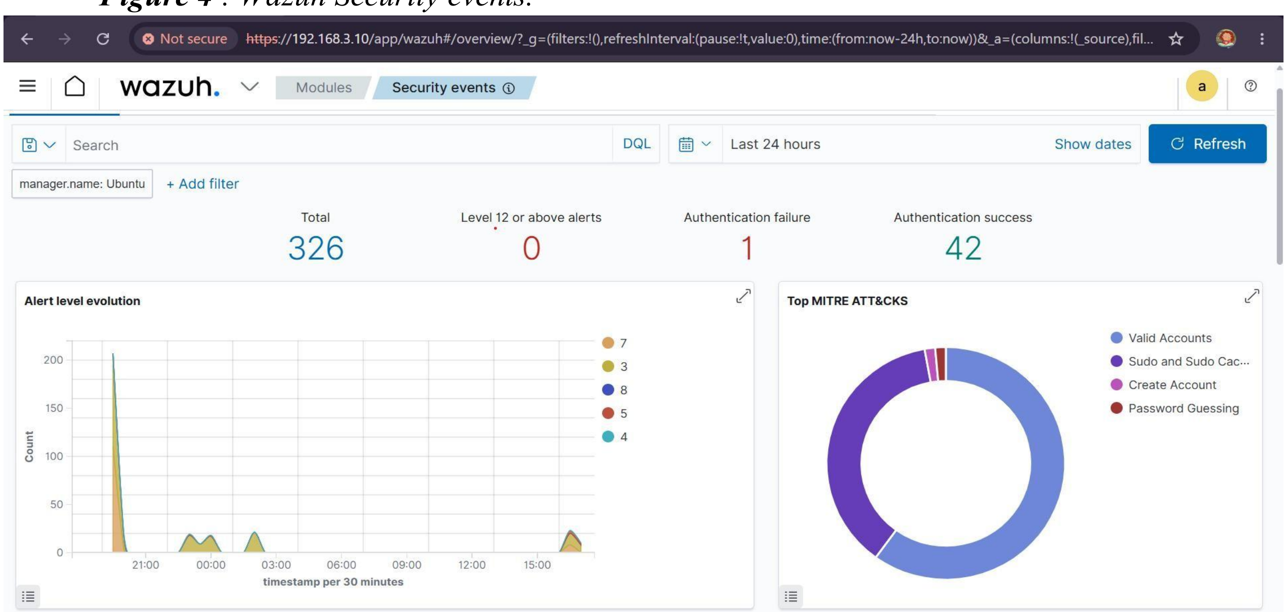


4. I monitored system logs and alerts through the Wazuh dashboard.

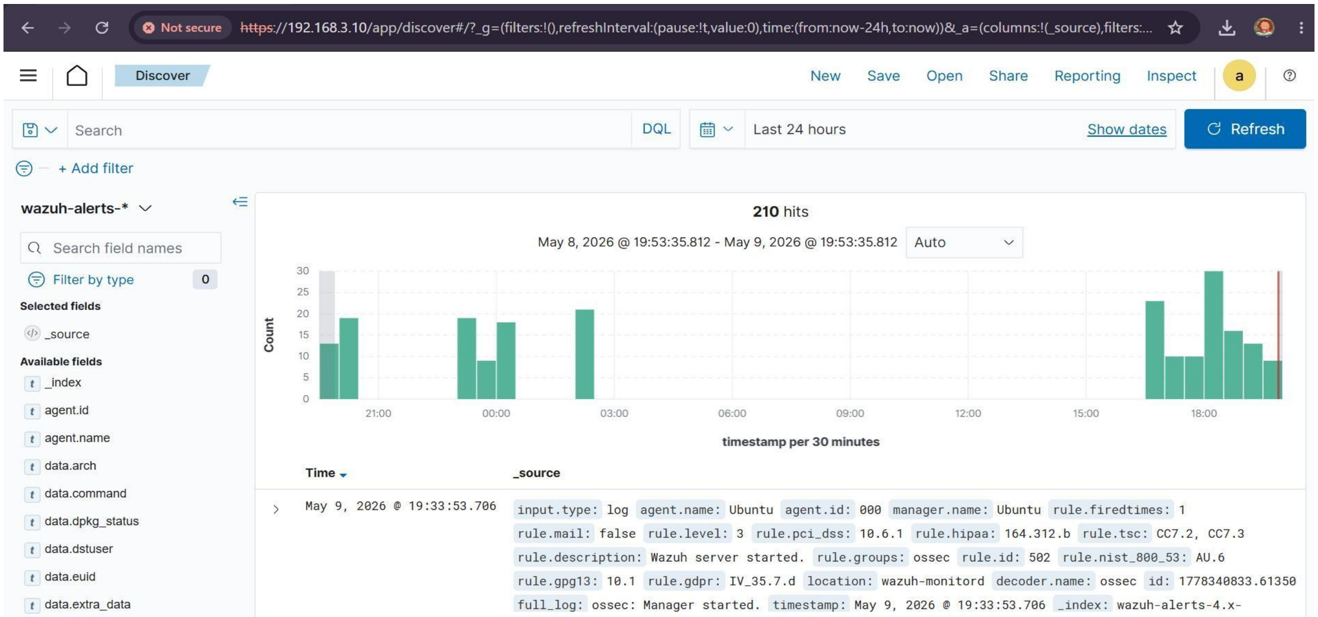
- *<u>Example Output :</u>*

- I saw unsuccessful login attempts in the logs.
- I received alerts for suspicious activities.
- I checked logs and alerts with the Wazuh dashboard.

- ***Figure 4** : Wazuh Security events.*



- ***Figure 5:*** *Security Alerts Detected by Wazuh.*



- ***Figure 6*** *: Terminal (alert).*

- *<u>SIEM Integration:</u>*

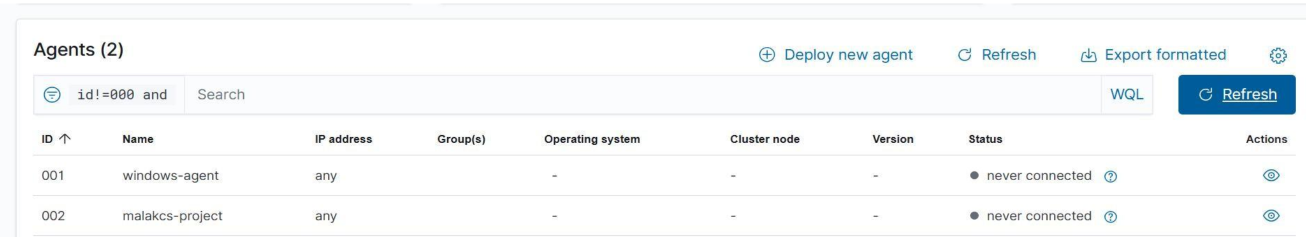
- I collected and analysed logs through a central system (Wazuh).
- I monitored and tracked system events in real-time, and triggered alerts for suspicious events.

- This shows how a SIEM system can be implemented to monitor and secure systems.

- ***Challenges:***

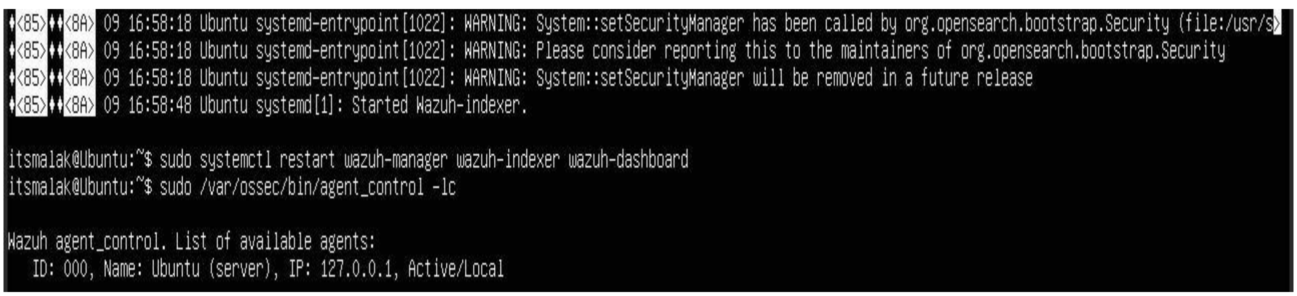
During the implementation, several challenges were encountered. The main issue was related to network communication between the agent and the manager, which caused the agent to appear as “never connected”. In addition, some logs did not immediately appear in the dashboard, requiring service restarts and configuration checks.

- ***Figure 7*** *: Wazuh agent status.*



- However, the local agent was successfully detected and shown as active within the system, as confirmed in the included screenshot.

- ***Figure 6*** *: Terminal (active local agent).*



- ***Conclusion:***

Through this project, I learned how SIEM systems help monitor and secure systems by collecting and analyzing logs in real time. I gained hands-on experience using Wazuh for detecting suspicious activities and generating security alerts.

- ***References :***

- Wazuh Documentation:https://documentation.wazuh.com
- Ubuntu Documentation:https://ubuntu.com/tutorials
- NIST Guide to Log Management: https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication800-92.pdf